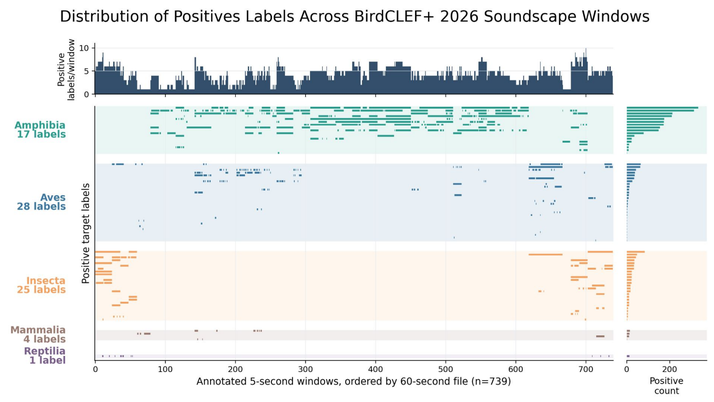
Transfer Learning for Avian Bioacoustics under Sparse Positive Labels

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Motivation

Passive acoustic monitoring makes continuous and non-invasive biodiversity assessment possible, but automated bird recognition is limited by **incomplete annotations**. Bioacoustic datasets offer valuable training signals, yet they differ in geographic coverage, species composition, and recording conditions, so **simply combining more data does not always improve performance**. This research project therefore investigates how **heterogeneous datasets** can be used as **weak positive supervision** and when transfer genuinely improves recognition. Our proposed framework combines ecological context, pretrained audio representations, temporal and label co-occurrence information, and **source-specific reliability modeling** to benefit from external and limit harmful transfer.

Datasets



BirdCLEF+ 2026 is our target benchmark, which contains **234 species labels** across 739 annotated 5-second windows. Its public annotations are **extremely sparse**: only **75 species have confirmed positive examples**, resulting in 3,122 positive window-label pairs and an overall label density of 1.81%. BirdCLEF 2021, iNatSounds, WABAD, and BirdSet PER/NES provide external recordings with different species coverage and recording conditions.

Table 1. Datasets used for target validation and transfer. BirdCLEF+ 2026 is the only evaluation target; all external data are mapped by normalized scientific name and used as positive-only transfer evidence.

Source	Overlap labels	Samples	Role
BirdCLEF+ 2026 [2]	234	739 windows	Target
BirdCLEF 2021 [3]	34	7,380 focal recordings	Transfer
iNatSounds [4]	183	4,002 train recordings	Transfer
WABAD [5]	96	8,887 windows	Transfer
BirdSet PER [6]	23	7,705 windows	Transfer
BirdSet NES [6]	19	1,223 positives	Transfer

Target Only Models

We first establish BirdCLEF-only baselines using ecological priors, lightweight acoustic features, and frozen AST, Whisper, and Wav2Vec2 embeddings. Ecological occurrence is represented by a smoothed site, month, and hour prior: $\widehat{p}_\ell(g) = \frac{n_{\ell,g} + \alpha \widehat{p}_\ell}{n_g + \alpha}$

where $n_{\ell,g}$ is the number of positive training windows for label ℓ in group g, n_g is the number of training windows in that group, $\widehat{p}_\ell$ is the global training prevalence of label ℓ, and α determines how strongly the group estimate is pulled toward this overall prevalence.

Positive-Unlabeled Transfer

To account for annotation uncertainty, we evaluate positive unlabeled (PU) weighting schemes that reduce the penalty assigned to unlabeled species-label pairs. The PU objective is

$$\mathcal{L}_{\mathrm{PU}} = -\sum_{i,\ell} [y_{i\ell} \log s_{i\ell} + \lambda(1 - y_{i\ell}) \log(1 - s_{i\ell})]$$

where $y_{i\ell}$ denotes the public positive indicator, $s_{i\ell}$ is the predicted score, and $0 < \lambda < 1$ reduces the contribution of unlabeled examples relative to confirmed positives.

Multi-Source Reliability Model

We combine target only and models associated with each external dataset that generate out-of-fold predictions with source coverage, ecological context, temporal position, species prevalence, and taxonomy:

$$\mathbf{x}_{i\ell} = [\text{model scores},\ \text{source coverage},\ \ \text{recording context},\ \text{species features}]$$

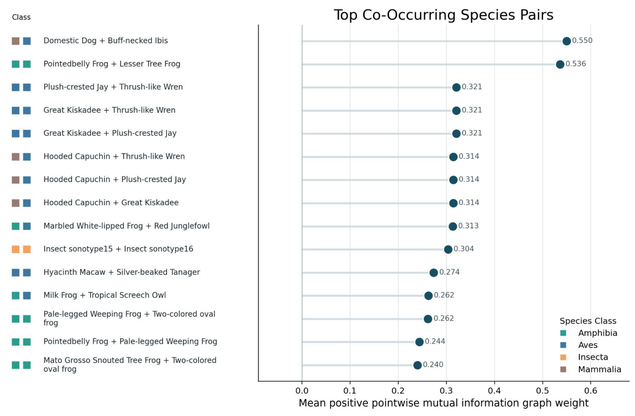
A regularized logistic meta classifier learns the final probability for species ℓ in window i: $q_{i\ell} = \sigma\left(\beta_0 + \boldsymbol{\beta}^\top \tilde{\mathbf{x}}_{i\ell}\right)$

This allows the model to learn when each external source should be trusted.

Label Relationships

A PPMI graph built from training data captures **species pairs that occur together** more often than expected from their individual frequencies.

$$\mathrm{PPMI}(\ell, m) = \max\left(0, \log \frac{\widehat{P}(\ell, m)}{\widehat{P}(\ell)\widehat{P}(m)}\right)$$



Evaluation

Models are evaluated using **5-fold cross validation**.We use **macro average precision** as the primary metric.

Model Performance Progression

We first isolate the contribution of each modeling component without using external datasets. **Ecological context** produces the largest improvement, increasing macro AP from 0.127 with a global prevalence prior to 0.462 with site, month, and hour information. Adding acoustic modeling, pretrained AST representations, and feature fusion further improves performance.

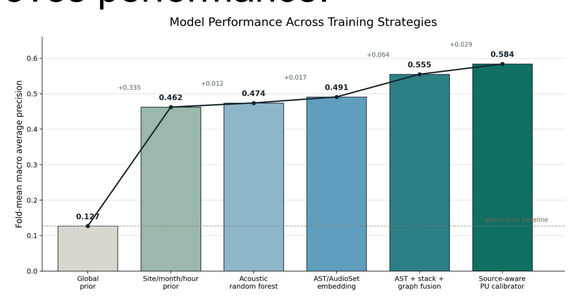


Fig. 2. Performance progression from ecological priors to multi-source reliability modeling on BirdCLEF+ 2026.

Multi-Source Transfer Results

Our experiments show that BirdSet PER and NES improve covered label performance but reduce all-label performance, indicating **negative transfer**. These results motivate the multi-source experiments below. We compare target-only, single-source, all-source, and leave-one-source configurations. Pooling every source achieves 0.578 macro AP, while excluding BirdSet PER produces the best fold-mean result of 0.584 macro AP and 0.860 macro AUC. This shows that **accounting for source reliability** is more effective than simply adding every available dataset.

Model	Macro AP	Macro AUC	Micro AP	Micro F1
AST+stack+graph fusion	0.555	0.832	0.670	0.361
AST+stack PU fusion	0.550	0.823	0.664	0.597
Markov-logit temporal smoother	0.559	0.837	0.675	0.364
Source-aware target-only, PU 0.2	0.572	0.826	0.629	0.608
Source-aware + BirdCLEF 2021 only, PU 0.2	0.568	0.834	0.631	0.603
Source-aware + iNatSounds only, PU 0.2	0.578	0.852	0.635	0.599
Source-aware + WABAD only, PU 0.2	0.576	0.852	0.630	0.608
Source-aware + BirdSet PER only, PU 0.2	0.557	0.829	0.628	0.603
Source-aware + BirdSet NES only, PU 0.2	0.549	0.830	0.625	0.607
Source-aware legacy sources, PU 0.2	0.583	0.858	0.639	0.602
Source-aware all sources, PU 0.2	0.578	0.859	0.639	0.601
Leave out BirdCLEF 2021, PU 0.2	0.580	0.859	0.636	0.597
Leave out iNatSounds, PU 0.2	0.572	0.853	0.628	0.602
Leave out WABAD, PU 0.2	0.573	0.850	0.634	0.601
Leave out BirdSet PER, PU 0.2	**0.584**	**0.860**	**0.640**	**0.602**
Leave out BirdSet NES, PU 0.2	0.583	0.858	0.639	0.600
Sequence on leave-out-PER source-aware model	0.555	0.838	0.654	0.611

Conclusion

Transfer learning under sparse annotations is a weak supervision and negative transfer problem. By preserving source identity and learning source reliability, our proposed framework achieves the best overall performance among the evaluated approaches.

Source code · github.com/AcaiLab/BirdMLSP

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